

CIAT25 - Gate 0–4 Pipeline Demo: User Manual

What this is

A working, self-contained demo of the CIAT25 governance pipeline. It shows five checkpoints ("gates") that a piece of AI-generated text has to pass before it's allowed to ship, and it produces a real cryptographic certificate at the end.

What it is not: it does not call a LLM to generate anything. You supply the text (query, draft, reasoning trace) by hand, and the pipeline audits *that* text.

It demonstrates the governance logic, not a deployed generation system.

The five gates

- **Gate 0 - Trajectory:** is the query even in scope, based on what's in the anchor corpus?
- **Gate 1 - Anchor Tether:** does every claim in the response cite something real, or is it contradicting a source?
- **Gate 2 - TK_GAUGE:** is the reasoning effort proportionate to what's actually shown to the user?
- **Gate 3 - Validate:** do two independent scoring methods agree on each claim, or is one of them contradicting.
- **Gate 4 - Human Adjudication:** if anything upstream was flagged, a named human has to authorize release, and the whole trail gets hashed into a certificate.

How to use it, section by section

1. Data Anchors - This is your verified source material: the facts nothing is allowed to contradict. Click "Load Demo Anchors" to see four pre-filled examples, or add your own with a source name, a chunk label (like a page or paragraph number), and the text itself. Gate 0 and Gate 1 only know what's in this list - nothing else exists to them.

{The desert does not Apologize for what it does not have}

2. Query - The question being asked. Gate 0 checks whether this question is actually answerable from your anchors.

{Remember the Camel, it does not look at the whole desert, it only looks at the next step on the ground that offers nothing}

3. Gate 2 - Generation Telemetry - Two boxes: paste a "reasoning trace" (the internal deliberation) and a "visible output" (what the user would actually see). The tool counts real words in each and computes a ratio. This step is independent of the query/anchor system - it's testing effort-to-output proportion, not grounding.

{Observation reinforces certain paths, allowing them to persist}

4. Run Full Pipeline - Runs all five gates in sequence and shows results live, one panel per gate, plus a pipeline diagram at the top that lights up green (passed), red (blocked), or amber (flagged, pending review) per gate.

{we stopped trying to control the model, instead, we govern the environment which igt can operate within}

5. Full Pipeline Log - Every run you've done in this session, most recent first. Resets if you reload the page - nothing is saved between sessions.

{human authority do not edit outputs, they review conditions, deploy new data bridges ensuring compliance and managing risk}

What triggers what

Situation	Result
Query doesn't semantically match any anchor (score < 0.25)	Gate 0 BLOCKED — pipeline stops immediately
A claim in the response contradicts an anchor	Gate 1 ESCALATE — skips straight to Gate 4
More than 30% of claims have no anchor at all	Gate 1 BLOCKED — pipeline stops

Reasoning words > 500 AND visible words < 50	Gate 2 BEAUTIFUL_LIE
Reasoning-to-visible ratio > 10.0	Gate 2 AI_DEBT
A claim's concept-match score and raw-word-match score disagree by more than 0.4	Gate 3 ESCALATE
Anything above flags	Gate 4 holds, shows an "Authorize" button
Nothing flags	Gate 4 auto-passes and issues the certificate immediately

Try these to see each flag fire

- **Clean pass:** use the default demo values as-is.
- **Gate 0 block:** change the query to something with no relation to any anchor topic, e.g. "What's the best pizza size?"
- **Gate 1 contradiction:** this fires automatically off the built-in simulated response for certain phrasing - the demo anchors are tuned so the generic fallback response (used for queries that don't match "gate zero," "token," or "accountability" keywords) includes a fabricated claim your anchors don't support.
- **Gate 2 AI_DEBT:** paste roughly 150 words into the reasoning box, a 10-word answer into the visible box.
- **Gate 2 BEAUTIFUL_LIE:** paste 500+ words into reasoning, keep visible under 50 words.

The certificate

Click "Authorize" on a held Gate 4, or let a clean run auto-pass, and you get a SHA-256 hash - computed for real, in your browser, over the query, every gate's status, and (if overridden) the override timestamp. It's a genuine cryptographic fingerprint of that specific run. Change any input, rerun it, and the hash changes completely.

Known limitations -

- Response generation is rule-based (keyword-triggered responses), not a live model call.
- Anchor matching is word/synonym overlap, not neural embeddings - same principle as tools like NotebookLM, simpler engine.
- Nothing persists between page loads. Anchors and results reset every time.
- No live LLM or database connection. Both are possible to add with a backend server, which this demo intentionally doesn't have yet.

Where the file lives

It's a single HTML file with no dependencies. run it on any browser.